

Experimental progress since the previous presentation

QUAD, ARCH-COMP, and four-way comparisons of time and width

Ours Huan Xiangru Native Flow*

September 29, 2026

Starting point: the 29-slide presentation of September 23, 2026.

All statuses come from saved evidence; no experiments were resumed to prepare this deck.

S00; latest native QUAD snapshot: September 29, 20:19:53 China time.

Main results from this phase

- **Broader coverage: 14 official configurations.** Seven completed in all four methods, each with five measured runs.
- **Our corrected P3 completed the full QUAD task.** All 1024 partitions were accepted through 1000 steps; corrected P2 had a common prefix of 799 steps.
- **Full P3 runtime fell by 51.36%.** It dropped from 52.56 to 25.56 minutes with identical saved ranges and final states.
- **Width interpretation changed materially.** The old native ACC reference has an enclosure failure; Unicycle still has a substantial width gap; Airplane now reaches a real NN property stop.

S01, S03–S06, S11–S12. The new QUAD candidate has not yet passed a full benchmark regression.

What the project computes

Given a set of initial states and a neural-network controller, we enclose every possible trajectory.

Stage	Role
NN bounds	Compute affine lower and upper bounds on the controller output over the current state set.
Dynamics propagation	Propagate the state using Taylor-model polynomials and interval remainders.
Validation and SR	Check inclusion of new remainders; SR retains cross-step error dependence.
Ranges and properties	Export endpoint and full-step tube bounds and evaluate the specified property.

Runtime measures cost; width measures conservatism. We must also check set inclusion and the portion of the task completed.

S01; the main comparisons include the closed-loop NN. Offline replays and one-step diagnostics are labeled separately.

Frozen revisions and explicit entry points

Method	Revision / implementation	Interpretation here
Ours	Huan-derived worktree fff9d0f; separate P2/P3 entry points	Not assumed merged into our public repository.
Huan	flowstar-gpu d5f0b68	Default mode differs from suite strict mode.
Xiangru	CROWN-Reach_Development 1c16d4e	Integrated driver, models, and GPU kernels.
Native Flow*	C++ library in CROWN-Reach 7b90f30	Fixed-step wrapper; repair labeled separately.

All three GPU suite rows use Xiangru's driver with strict dynamics and box/same-slope/RPC32. Independent pendulum defaults use hybrid/parity.

Huan and Xiangru share 27/29 core Python files and the same CUDA math; 13 comparable range files are byte-identical.

S01. These comparisons do not characterize every author branch or later revision; exact repository identities are in the data bundle.

How to read runtime and width

Measure	Definition and scope
Measured runtime	Wall time of a fresh process, including imports, cached extension loading, NN initialization, and solve; median of five runs; excludes first compilation.
Single diagnostic	May include observation, checkpoints, and audits; time before failure is never used as a full-task speed denominator.
Per-partition width	Upper minus lower bound for one partition, coordinate, and time; report mean or maximum.
Global hull width	Largest upper minus smallest lower bound over partitions: $W_{\text{hull}} = \max_b u_b - \min_b l_b$.
Width ratio	GPU / native, only over a common valid prefix. Zero or tiny denominators are identified separately.

An endpoint is the end of a step; a tube covers the whole step. Mean partition width and global hull width are different measures.

S01; runs shared a V100 node rather than an exclusive machine.

Coverage of the 14 official configurations

Group	Configurations	Outcome
Four-way complete: 7	Attitude, Unicycle, two heterogeneous TORA variants, NAV robust, Single Pendulum, ACC*	Five full runtimes per method; a defect in the ACC reference was found later.
Partial: 4	DP less robust, NAV standard, CartPole, TORA homogeneous	Native timeouts or GPU rejections; retain common prefixes.
Property stops: 2	DP more robust, Airplane	Stop semantics and actual model must be checked; Airplane later ran with a GPU NN connection.
Targeted repair: 1	QUAD	Earlier numerical failure or resource stop; P3 now completes 1000 steps.

VCAS provides an NN service/simulation, but no comparable C++ reachability entry point was found; it is not counted as a fifteenth case.

S01, S11. Completion of a run does not by itself establish a full rigorous NNCS proof.

Seven complete cases: four-way runtime

Benchmark	Ours	Huan	Xiangru	Flow*
Attitude	7.121	6.990	7.028	6.315
Unicycle	6.909	9.796	9.581	10.722
TORA ReLU/tanh	7.685	12.421	12.332	8.632
TORA sigmoid	8.208	13.165	13.274	9.098
NAV robust	9.730	14.646	14.814	68.164
Single Pendulum	5.176	5.597	5.618	4.908
ACC*	7.793	8.035	7.960	8.259

Seconds; median of five complete fresh-process runs. Ours is about $7.01\times$ faster on NAV robust and $1.55\times$ faster on Unicycle.

Attitude and Single Pendulum are still slower than native. ACC* retains only the historical execution time.

S01. The old native ACC* reference was later shown to miss reachable states; it cannot certify accuracy against a correct reference.

Complete cases: worst width ratio over time

Benchmark	Huan local	Xiangru local	Ours local	Ours global
Attitude	1.1456	1.1456	1.1456	1.1456
Unicycle	3.4435	3.4435	3.4994	3.4994
TORA ReLU/tanh	1.0077	1.0077	1.0081	1.0081
TORA sigmoid	1.0107	1.0107	1.0107	1.0107
NAV robust	1.0388	1.0388	1.0465	1.0123
Single Pendulum	1.0921	1.0921	1.0581	1.0581

Each entry is the worst ratio to native across common times, coordinates, and endpoint/tube views.

Heterogeneous TORA and NAV robust are close to native. The roughly $3.50\times$ Unicycle gap reflects real width, not rounding in the last digit.

S01. Local means per partition; global means the union hull. Old ACC ratios are corrected separately and excluded from the accuracy ranking.

Unicycle: faster, but a wider final range

Endpoint at T=10	Flow*	Huan	Xiangru	Ours
x_1	0.04038534	0.09540937	0.09540937	0.09669186
x_2	0.06082288	0.1409442	0.1409442	0.1428876
x_3	0.04165892	0.1313937	0.1313937	0.1337685
x_4	0.03520954	0.1212424	0.1212424	0.123212

Five-run median runtime: ours 6.909 s; Huan 9.796 s; Xiangru 9.581 s; Flow* 10.722 s.

Global width in coordinate 4 is 0.03520954 versus 0.12321200: a $3.499\times$ gap.

S01. B1, $h=.02$, solution order 2, 500 steps; endpoint global hull widths are shown.

Unicycle: narrowing down the first divergence

- Fifty native NN replays agree on the same input; closed-loop input boxes may still differ between implementations at each control period.
- For the same GPU first-step polynomial, separate bounding of the x_3/x_4 tubes loses shared time dependence and explains part of the extra width. Whether native uses the same internal polynomial is unproven.
- The additional difference between ours and Huan/Xiangru appears within the first control period; its internal-state cause remains unresolved.
- Five native TM/SR boundaries from the first step were reproduced and saved. Initialization also under-enclosed one lower endpoint by 1 ULP.
- The injected GPU first-step diagnostic stopped during loading and yielded no new numerical result.

S12. The 1-ULP initial gap alone cannot explain the final 3.5-fold width difference.

NAV: global width is close; standard native run remains incomplete

Configuration	Runtime and completion	Width
NAV robust, B25	All four complete 600 steps. Ours 9.730 s; Huan 14.646 s; Xiangru 14.814 s; native 68.164 s.	Our worst global ratio is 1.0123; worst per-partition ratio is 1.0465.
NAV standard, B640	All three GPUs complete 600 steps. Four-worker native solve times out at 300 s, with only 220 steps saved.	On the common 220 steps, worst global ratio is 1.0096; local maximum is 17.0645.

The 17.0645 ratio occurs at step 59, lane 626, x_3 :

native 3.44169×10^{-15} , ours 5.87308×10^{-14} .

It does not mean the entire reachable set is 17 times wider.

S01. Robust timings are five-run medians; standard results are single observed diagnostics and must not be pooled with them.

Airplane: from build failure to a real NN closed loop

Implementation / stage	Accepted steps	Process s	Stop reason
Huan frozen engine, shared driver	0	35.482	Build RSS guard
Xiangru frozen engine, shared driver	0	35.793	Build RSS guard
Our sparse NNCS	79	46.628	Property stop
Ours with time bisection check	78	45.313	Property stop
Native Flow*	78	8.465	Property stop

Eight actual CROWN calls are connected. Sparse metadata follows the true monomial support while preserving P6 and the original task.

The target is 200 steps; these single stopped runs do not yield a complete four-way speed ranking.

S01, S11. Huan/Xiangru are the main-suite strict entries, not replacements using our sparse port.

Airplane: a concrete witness for stopping and width

At step 78, evaluate $y - 1 \leq 0$ on the same native accepted model:

Time interval	Native constraint interval	Decision
[0, .01]	[−.00049283, .00902177]	Crosses zero
[.005, .01]	[.00423059, .00898790]	Entirely positive: violates property

The actual GPU polynomial also has an independent Fraction witness. Bounded bisection aligns the stop to step 78 instead of 79 in this case.

Step-78 y tube width: ours 0.0095823622; native 0.0095145997.

Worst tube ratio over the common 78 steps is 1.027216; tiny-denominator endpoint ratios are treated separately.

S11. The range prefix is byte-identical; this case does not establish identical stopping semantics on every system.

ACC: the narrower native reference needs correction

The frozen native library under-encloses the ACC lead-vehicle subsystem.

For $v(0) = 32$, $a(0) = 0$, $t = 0.2$:

Source	Interval for acceleration a
Independent analytic trajectory enclosure	$[-0.6762395216, -0.6761530739]$
Old native exported endpoint	$[-0.6766724592, -0.6763896330]$

The intervals are disjoint. Both direct native ODE execution and the matched wrapper reproduce the issue; it lies in a refinement VAR truncation tail.

An isolated patch retains the required tail; all 50 steps and the analytic witness then pass.

S04. This finding concerns the archived library, not every Flow* revision; disabling refinement is not presented as the repair.

ACC: widths against the repaired reference

Implementation	Step-2 a_{lead} endpoint width	Recorded span
Old native (under-enclosed)	0.00028282615	50 steps
Native, VAR-tail repaired	0.00224735439	50 steps
Huan strict	0.00224700883	50 steps
Xiangru strict	0.00224700883	50 steps
Our control	0.00224556233	50 steps
Our validation-order candidate	0.00077213783	50 steps

The candidate retains solution order 3 and changes only RHS validation order; this endpoint is 65.6% narrower than our control.

Single observed processes: repaired native 8.2968 s; candidate 8.4740 s. The candidate's worst tube ratio to repaired native is still about 1.40993.

S04. These single observed diagnostics do not establish a speed gain; the candidate is not the suite-wide default.

QUAD: fixed task and explicit algorithm variants

Fixed task

Variants in this phase

12 physical states, 16 total variables; $x_{1:6} \in [-.4, .4]$, others zero; 1024 actual partitions.

$h = .005$, $T = 5$: 1000 ODE steps; control period .1, 50 NN calls.

Cutoff 10^{-6} , remainder cap $\pm .1$, SR capacity 1000, full history retained.

Current matched-box comparisons use the 8dd initial array / native-f64. The historical ARCH table used f90 arrays / RPC32; last-bit endpoint differences are disclosed separately.

S02–S03, S05. Partition grid: $8 \times 8 \times 8 \times 2 \times 1 \times 1$; the official physical-model expressions are retained.

Huan's fast QUAD result was reproduced

Mode / implementation	Common steps	Seconds	Record type
Huan parity + chunked SR	1000	75.250	Five full-process median
Huan strict, same 8dd/f64	596	33.765	Failed-process total
Xiangru ARCH strict	596	61.564	Failed-process total
Our corrected P2	799	658.477	Failed watch total
Our P3 + trig reuse	1000	1533.752	Single full watch

The author's saved log reports 83.048 s; independent reproduction confirms that parity can complete quickly.

Parity and strict have different error contracts. Current P3 is much slower than Huan's fast mode, and a complete strict four-way ranking is still unavailable.

S02–S06. Completion, timing boundaries, and modes differ across rows; no speedup is computed across this table.

QUAD: where completion gains came from

Area	Change	Observed extent
Mathematics and bounds	Repair reciprocal remainder and end-point/injection rounding charges; retain SR relations; regroup full factors every 20 steps.	Corrected P2 reached a full-partition prefix of 799 steps.
Higher work order	Explicit P3 retains more polynomial information; validation order also increases.	Completed 1000 steps after a higher resource budget.
Memory engineering	Keep complete factors on CPU and rebuild GPU chunks; manage graph and cache lifetime.	No SR history, partitions, or time horizon removed.
Avoid duplicate work	Reuse identical $\sin/\cos(\text{var})$ results and caches within one validation.	Halved full P3 time with identical saved output.

S03, S05–S06, S13. P2 rejection at step 800 and the old P3 GPU-memory guard are distinct problems.

QUAD at step 580: widths on the same initial boxes

State	Huan strict	Our P2	Our P3
x1	2.950594	2.207218	1.822787
x3	0.8007371	0.1900823	0.09226701
x5	4.519631	2.002716	1.40131
x6	2.62794	0.4687641	0.2156717
x7	0.4023827	0.06399812	0.0187245
x11	4.457102	0.6060088	0.261089

At $t = 2.9$, all 1024 partitions are accepted. Entries are **mean per-partition endpoint widths**.

Huan and our variants differ in order and boundary handling, so this is a method comparison. No native full-partition output or matched Xiangru data exist here.

S07. Only Huan mean/max are available, not a global hull; corresponding new-trig widths equal P3 widths.

QUAD at step 799: width across P2/P3 variants

State	P2 mean	P3 mean	P3 / P2
x1	4.380172	2.835693	0.6474
x3	1.033031	0.07780014	0.0753
x5	5.822284	1.586771	0.2725
x6	3.869302	0.1965135	0.0508
x7	0.5352477	0.01414275	0.0264
x11	6.868202	0.1692858	0.0246

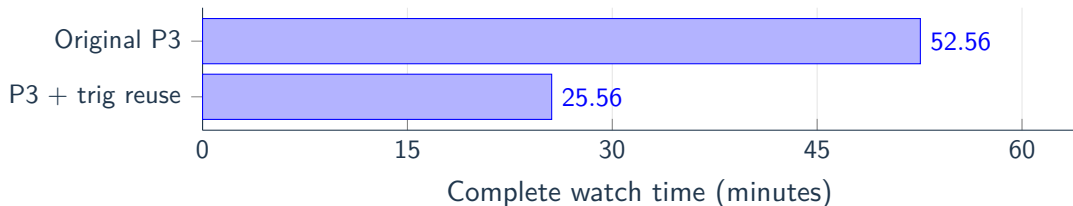
Common valid time $t = 3.995$; mean per-partition endpoint width over 1024 partitions.

P2 first rejects 12 partitions at step 800; P3 eventually completes all 1000 steps.

These are different orders and numerical variants; higher order is not guaranteed to be faster or successful.

S03, S05. Partitions surviving P2 after step 800 cannot represent the full original initial set.

Full QUAD P3: 52.56 down to 25.56 minutes



Timing component	Original P3 s	Reuse P3 s	Original / new
Complete watch	3153.449456	1533.752052	2.056
Cumulative advance	2868.586754	1234.852041	2.323
Cumulative NN calls	3.420546	3.533022	0.968

Same full 1000-step configuration: 51.36% less time in one run, not a five-run benchmark timing.

S06. Includes initialization, graph setup, logging, and checks; a native job was also running on other GPU/CPU resources.

QUAD speedup preserved every saved numerical output

- **All 1000 observer files are identical.** They include endpoint/tube ranges, acceptance masks, and states for all 1024 partitions.
- **All 50 controller, 50 transfer, and terminal files are identical.**
- **Two final complete states match.** Plant, 14 SR/host components, progress, and final SR reset were checked.
- The model, initial set, P3 orders, remainder cap, SR history, and 1000-step workload were retained.
- Peak GPU memory was about 11.15 GiB and RSS about 5.63 GiB; GPU resource guard was 14 GiB.

S06. Actual remote CPU data were checked and key JSON records archived locally; these provide evidence of output equivalence for the recorded runs.

QUAD at T=5: mode differences still matter

State	Huan parity global hull	Our P3 global hull
x1	6.4997094	6.6236214
x2	7.0530187	7.2535537
x3	0.047687232	0.068972382
x5	1.6005968	1.7220725
x6	0.12335096	0.1799271
x7	0.0070165625	0.01085314

Both complete to T=5, but parity and corrected P3 differ in order, error handling, and bounds.

The new trig and original P3 have identical widths. Full T=5 values for Huan strict, Xiangru strict, and Flow* are still unavailable.

S02, S05–S06. Absolute values are juxtaposed as observed; a narrower enclosure alone does not prove correct coverage.

QUAD versus the new native entry: common 40 steps

State	Our mean	Native mean	Our global hull	Native global hull
x1	0.1806475	0.1834987	0.960742	0.963601
x3	0.2144799	0.2165988	0.8208428	0.8229604
x5	0.8112718	0.8207659	0.8128924	0.8221873
x6	0.6027428	0.6030136	1.477771	1.478059
x7	0.00461459	0.004584761	0.005191669	0.005161935

At $t = .2$, both accept all 1024 partitions on the same requested boxes / native-f64.

Our method is P3; native is P2 with repaired initial coverage. Actual affine forms and subsequent NN domains need not match bitwise.

Our/native observed 40-step process times are 155.452 / 171.138 s; diagnostic overhead differs, so no benchmark speedup is reported.

S08, S10. Our time here is the pre-optimization P3 diagnostic, not the full trig-reuse time or a 40-step benchmark timing.

Native QUAD: initial-coverage repair and 400-step snapshot

- On the same 8dd requested boxes, native initialization under-enclosed 384 lower endpoints across 338 partitions by 1 ULP.
- An isolated variant retains the original center and library while taking the maximum outward distance as its radius; all 16,384 exact coverage checks pass.
- After the repair, all 40 steps and two NN calls are accepted. Replays of the two actual NN inputs match 86,016 T/L/U floating-point outputs.
- **Last saved snapshot: 400/1000 steps; all 1024 partitions accepted.**
Timestamp: September 29, 20:19:53 China time. No terminal state is available.
- Host-memory budget: 128 GiB; GPU guard: 14 GiB; four CPU workers. No complete T=5 time or width is available.

S10. This report reads a local saved snapshot without reconnecting to the server; 400 steps is not a live status.

What the correctness checks support

Evidence obtained

Exact initial-set inclusion; replayed interface ordering and dtype.

Cold resume; accepted steps and partitions; numerical and SR component comparisons.

ACC analytic counterexample and isolated fix; generic reciprocal counterexamples located.

Parameters, resource changes, failures, and incomplete runs recorded.

Still outside its scope

End-to-end proof of controller FP bounds and RN injections.

All inputs, combinations of operations, and future trajectories.

Generic counterexamples do not prove lost trajectories in every benchmark.

Suite regression and five timed runs for the new QUAD candidate.

No evidence of reduced work, discarded required error, or hidden rejections.

A “strict” switch, VERIFIED log, or similar width does not prove end-to-end correctness.

S01, S04, S06, S10, S12–S13.

Current progress and next priorities

Established so far

Four-way screening of 14 configurations; seven cases with five timed runs per method.

Complete P3 QUAD; same outputs in roughly half the total time.

Airplane with actual NN calls and an explained stop at step 78.

Corrected ACC reference and narrowed localization of Unicycle width gap.

To complete later

Rerun affected regressions for new corrected revisions, rather than inherit older results.

Match modes for Huan/Xiangru and complete native T=5; then repeat timing.

Huan/Xiangru metadata-resource issue and cost under a common stop contract.

General regression of the repaired reference, remaining tube gap, and Unicycle internal attribution.

This presentation summarizes saved experiments only; the experimental task remains paused.

Numbers are current through the saved evidence; the listed future work has not yet been carried out.

Appendix: Attempts without four complete results

Case	Ours	Huan	Xiangru	Native Flow*
DP more robust	early stop 5.36	early stop 5.63	early stop 5.76	early stop 5.11
DP less robust	complete 8.76	complete 8.30	complete 8.51	timeout 303.66
NAV standard	complete 14.15	complete 18.12	complete 17.91	timeout 303.93
CartPole	rejected 12.65	rejected 13.77	rejected 13.99	numeric stop 8.94
TORA homogeneous	complete 7.88	rejected 8.90	rejected 8.82	complete 8.20
QUAD	rejected 35.81	rejected 58.76	rejected 61.56	RSS stop 845.82

Seconds; one observer-instrumented run per method; separate from five-run medians.

QUAD uses the initial ARCH contract; later variants are labeled. See Airplane in the main slides.

DP more robust actually loaded the less robust model; both labels are recorded.

S01. Full step counts, outcomes, and resources: `arch_diagnostic_attempts.csv`.

Appendix: Official benchmark settings (1/2)

Case	B	h / order	ctrl. $\times$ N	T	steps	rem.
Attitude	1	0.05/3	0.1×30	3	60	± 0.01
DP more robust	1	0.005/4	0.02×20	0.4	80	± 0.01
DP less robust	225	0.01/4	0.05×20	1	100	± 0.01
NAV robust	25	0.01/4	0.2×30	6	600	± 0.1
NAV standard	640	0.01/4	0.2×30	6	600	± 0.1
Unicycle	1	0.02/2	0.2×50	10	500	± 0.01
TORA ReLU/tanh	1	0.01/6	0.5×10	5	500	± 0.01

Cutoff 10^{-6} ; default SR 1000, except ACC at 50. Unicycle also has disturbance $w \in [-10^{-4}, 10^{-4}]$. B is the number of initial boxes.

This is the historical ARCH contract. The current QUAD P3 order and native-f64 variant are specified separately.

S01. Exact initial boxes, splits, model/box hashes, and configuration files: `data/arch_configurations.csv`.

Appendix: Official benchmark settings (2/2)

Case	B	h / order	ctrl. $\times$ N	T	steps	rem.
TORA sigmoid	1	0.01/6	0.5×10	5	500	± 0.01
CartPole	1	0.005/6	0.02×50	1	200	± 0.1
Airplane	1	0.01/6	0.1×20	2	200	± 0.01
Single Pendulum	1	0.01/2	0.05×20	1	100	± 0.01
TORA homogeneous	12	0.1/3	1.0×20	20	200	± 0.01
ACC*	1	0.1/3	0.1×50	5	50	± 0.1
QUAD	1024	0.005/2	0.1×50	5	1000	± 0.1

Cutoff 10^{-6} ; default SR 1000, except ACC at 50. Unicycle also has disturbance $w \in [-10^{-4}, 10^{-4}]$. B is the number of initial boxes.

This is the historical ARCH contract. The current QUAD P3 order and native-f64 variant are specified separately.

S01. Exact initial boxes, splits, model/box hashes, and configuration files: `data/arch_configurations.csv`.

Appendix: Runtime sample ranges (1/2)

Case	Ours	Huan	Xiangru	Flow*
Attitude	7.121 [6.979, 7.186]	6.990 [6.960, 7.059]	7.028 [6.839, 7.376]	6.315 [6.111, 6.355]
Unicycle	6.909 [6.852, 7.042]	9.796 [9.622, 10.058]	9.581 [9.544, 10.054]	10.722 [10.554, 10.778]
TORA ReLU/tanh	7.685 [7.659, 7.875]	12.421 [12.137, 12.503]	12.332 [12.226, 12.705]	8.632 [8.617, 8.676]
TORA sigmoid	8.208 [8.127, 8.310]	13.165 [12.969, 13.259]	13.274 [13.188, 13.515]	9.098 [9.034, 9.121]

Each cell gives median [min, max] in seconds across five runs.

V100 GPU 3 and CPUs 14–17 were fixed; runs were interleaved and each group's first warmup was excluded. The node was shared.

S01. The 140 counted runs and 28 excluded warmups: arch_timing_samples.csv; see main slides for the ACC correction.

Appendix: Runtime sample ranges (2/2)

Case	Ours	Huan	Xiangru	Flow*
NAV robust	9.730 [9.349, 9.967]	14.646 [14.071, 15.074]	14.814 [14.045, 14.916]	68.164 [66.916, 78.318]
Single Pendulum	5.176 [5.068, 5.375]	5.597 [5.527, 5.624]	5.618 [5.504, 5.819]	4.908 [4.773, 4.917]
ACC*	7.793 [7.740, 7.868]	8.035 [7.876, 8.078]	7.960 [7.862, 8.081]	8.259 [8.224, 8.342]

Each cell gives median [min, max] in seconds across five runs.

V100 GPU 3 and CPUs 14–17 were fixed; runs were interleaved and each group's first warmup was excluded. The node was shared.

S01. The 140 counted runs and 28 excluded warmups: `arch_timing_samples.csv`; see main slides for the ACC correction.

Appendix: Endpoint widths — Attitude and Pendulum

State	Flow*	Huan	Xiangru	Ours
Attitude x_1	0.004194134	0.004174214	0.004174214	0.004398305
Attitude x_2	0.005980154	0.005935659	0.005935659	0.006299372
Attitude x_3	0.006193977	0.006172913	0.006172913	0.006555204
Attitude x_4	0.03229213	0.03207033	0.03207033	0.03384848
Attitude x_5	0.01763594	0.01736152	0.01736152	0.01848375
Attitude x_6	0.0186374	0.01847936	0.01847936	0.01971919
Single Pendulum x_1	0.1391394	0.1482617	0.1482617	0.1438742
Single Pendulum x_2	0.1444355	0.1551737	0.1551737	0.1502648

Endpoint width of the global hull at the final time; state units follow each model.

S01. CSV retains full precision and per-state statistics over time; see S04 for the corrected ACC reference.

Appendix: Endpoint widths — two heterogeneous TORA cases

State	Flow*	Huan	Xiangru	Ours
TORA ReLU/tanh x_1	0.02512807	0.02507951	0.02507951	0.02509966
TORA ReLU/tanh x_2	0.02745754	0.02741052	0.02741052	0.02739115
TORA ReLU/tanh x_3	0.02203482	0.02199292	0.02199292	0.02201384
TORA ReLU/tanh x_4	0.02161695	0.02159111	0.02159111	0.02165234
TORA sigmoid x_1	0.02487467	0.02480363	0.02480363	0.02489772
TORA sigmoid x_2	0.02635339	0.02628228	0.02628228	0.02628759
TORA sigmoid x_3	0.02610609	0.02602511	0.02602511	0.02608528
TORA sigmoid x_4	0.02799618	0.02794208	0.02794208	0.02804373

Endpoint width of the global hull at the final time; state units follow each model.

S01. CSV retains full precision and per-state statistics over time; see S04 for the corrected ACC reference.

Appendix: Endpoint widths — NAV robust and Unicycle

State	Flow*	Huan	Xiangru	Ours
NAV robust x_1	0.0909985	0.09030302	0.09030302	0.09134372
NAV robust x_2	0.01507649	0.01480299	0.01480299	0.01511633
NAV robust x_3	0.02489178	0.02461458	0.02461458	0.02495127
NAV robust x_4	0.2941089	0.2924141	0.2924141	0.294786
Unicycle x_1	0.04038534	0.09540937	0.09540937	0.09669186
Unicycle x_2	0.06082288	0.1409442	0.1409442	0.1428876
Unicycle x_3	0.04165892	0.1313937	0.1313937	0.1337685
Unicycle x_4	0.03520954	0.1212424	0.1212424	0.123212

Endpoint width of the global hull at the final time; state units follow each model.

S01. CSV retains full precision and per-state statistics over time; see S04 for the corrected ACC reference.

Appendix: Endpoint widths — historical ACC output

State	Flow*	Huan	Xiangru	Ours
ACC* x_1	20.99393	21.01204	21.01204	21.01046
ACC* x_2	0.1975723	0.2017539	0.2017539	0.2016287
ACC* x_3	0.0006846514	0.0007410763	0.0007410763	0.0007376998
ACC* x_4	5.028369	5.497976	5.497976	5.333353
ACC* x_5	1.795632	2.015511	2.015511	1.948656
ACC* x_6	1.0637	1.181106	1.181106	1.15943

Endpoint width of the global hull at the final time; state units follow the model.

The historical native ACC result omitted reachable states. This table preserves its output and is not a valid accuracy reference.

S01. CSV retains full precision and per-state statistics over time; see S04 for the corrected ACC reference.

Appendix: QUAD widths at T=5, states 1–6

State	P3 mean	P3 max	P3 hull	Huan parity hull
x1	3.8904615	3.9143939	6.6236214	6.4997094
x2	6.5097634	6.5323884	7.2535537	7.0530187
x3	0.062617019	0.064438385	0.068972382	0.047687232
x4	1.090248	1.1008754	1.5039563	1.4226317
x5	1.7028656	1.7126352	1.7220725	1.6005968
x6	0.1679	0.17312141	0.1799271	0.12335096

Original P3 and the trig-reuse version have identical saved numbers. Huan parity and P3 use different error and order settings.

x12 is constant; P3's roughly 4.45×10^{-308} is roundoff noise, so no ratio is reported.

S02, S05–S06. Complete T=5 widths are unavailable for strict Huan, strict Xiangru, and Flow*; no values were imputed.

Appendix: QUAD widths at T=5, states 7–12

State	P3 mean	P3 max	P3 hull	Huan parity hull
x7	0.010151003	0.010497058	0.01085314	0.0070165625
x8	0.0092057392	0.0095347946	0.0097108278	0.0061199043
x9	0.0060247288	0.0061434876	0.0061434876	0.0044494068
x10	0.14977017	0.15565646	0.15565646	0.086179577
x11	0.12566507	0.13084117	0.13084117	0.071020671
x12	$4.4501477 \times 10^{-308}$	$4.4501477 \times 10^{-308}$	$4.4501477 \times 10^{-308}$	0

Original P3 and the trig-reuse version have identical saved numbers. Huan parity and P3 use different error and order settings.

x12 is constant; P3's roughly 4.45×10^{-308} is roundoff noise, so no ratio is reported.

S02, S05–S06. Complete T=5 widths are unavailable for strict Huan, strict Xiangru, and Flow*; no values were imputed.

Appendix: QUAD four-way reference at t=0.5, states 1–6

State	Our P3	Our P2	Huan	Xiangru	Flow*
x1	1.209483	1.210965	1.212697	1.212697	1.215011
x2	1.208691	1.211738	1.216197	1.216197	1.21431
x3	0.4934754	0.4983923	0.5015186	0.5015186	0.4962256
x4	0.8259578	0.8302554	0.8346602	0.8346602	0.8348579
x5	0.8578122	0.8743562	0.8928963	0.8928963	0.8676566
x6	1.526041	1.533748	1.538956	1.538956	1.529488

Endpoint global hull for the common valid prefix of 100 steps.

The three ARCH runs use f90 initial boxes / RPC32. Our P2/P3 use 8dd / native-f64, with different orders.

This is a descriptive reference with contract differences, not a strict same-input algorithm ranking.

S09. The named CSV and RESULT files give all state widths, contract differences, and provenance.

Appendix: QUAD four-way reference at t=0.5, states 7–11

State	Our P3	Our P2	Huan	Xiangru	Flow*
x7	0.01237843	0.01484585	0.01747069	0.01747069	0.01240678
x8	0.01166923	0.01405786	0.0165278	0.0165278	0.01170151
x9	0.0001417913	0.0001689534	0.0002153181	0.0002153181	0.0001298099
x10	0.2100149	0.2139019	0.2186553	0.2186553	0.2102979
x11	0.1978412	0.2015494	0.2059809	0.2059809	0.1981034

Endpoint global hull for the common valid prefix of 100 steps.

The three ARCH runs use f90 initial boxes / RPC32. Our P2/P3 use 8dd / native-f64, with different orders.

This is a descriptive reference with contract differences, not a strict same-input algorithm ranking.

S09. The named CSV and RESULT files give all state widths, contract differences, and provenance.

Appendix: Fixed-plant baselines reported previously

Case	Ours	Huan	Xiangru	Flow*
Van der Pol B1	2.921487	12.552073	12.602979	1.447977
Van der Pol B32	0.769654	0.840225	0.837461	0.682880
Brusselator B1	4.456490	16.867330	16.737087	13.847315
Brusselator B32	0.897973	0.929344	0.935446	8.053289

These are five-run medians of core execution time from the previous slides; they were not rerun here.

Core time excludes new-process startup and separate export. It cannot be ranked directly against the new NNCS process wall time.

S00. Original VDP with one box remains slower than native Flow*; current work centers on official NNCS cases and QUAD.

Appendix: Evidence index and version scope

Source ID	Contents
S00 / S01	Prior slides/README; 14-case report, timings, widths, settings.
S02 / S03	Huan QUAD parity/strict runs; corrected P2 to step 799.
S04 / S05	ACC counterexample/fix; full P3 and step-799 comparison.
S06	Full 1000-step trig run; INPUT/RESULT/COMMAND; CPU audit.
S07 / S08 / S09	QUAD widths at steps 580, 40, 100.
S10 / S11 / S12 / S13	Native coverage/snapshot; Airplane; Unicycle; operations/failures.

SOURCES.json maps original and copied paths with SHA-256 hashes.

Missing results remain blank or have an explicit status.

Sources and URLs: SOURCES.md. Large PT/SR files, weights, and environments remain on the server.